Week 13: Advanced Risks: Goal Misgeneralization, Power-Seeking and Deception

Machine Ethics and Artificial Intelligence Safety

Machine Ethics and Artificial Intelligence Safety (11118BLG001). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Some risks become serious only as systems grow more capable and autonomous. This week examines learned goals that differ from intended ones and theoretical reasons to expect power-seeking. It also covers the problem of keeping systems correctable, recent empirical studies of deceptive behaviour and the evaluations used to detect such behaviour [1, 2, 3, 4].

Learning outcomes

By the end of the week, students are expected to distinguish outer from inner alignment and to explain goal misgeneralization with an example. They should be able to state the instrumental convergence argument and its formal version and to analyse the off-switch game. Finally, they should be able to evaluate empirical claims about deceptive behaviour by separating capability from propensity.

Lecture

Inner alignment and goal misgeneralization

Earlier weeks treated misalignment as a problem of specification, called outer alignment. Hubinger and colleagues pointed to a second problem [1]. A learned model can itself implement an optimiser, called a mesa-optimiser, whose objective differs from the training objective even when that objective is correct. Ensuring that the learned objective matches the intended one is the inner alignment problem.

Goal misgeneralization is an observable form of this concern. Langosco and colleagues trained agents in a video game in which the coin always appeared at the end of the level [5]. When the coin was moved at test time, the agents still ran to the end of the level: Their navigation skills generalised, but their goal did not. Shah and colleagues collected further examples and stressed

that the failure can occur even when the specification is correct, because many goals are consistent with the same training data [6]. A system that is competent in pursuit of the wrong goal can be more dangerous than one that simply fails. The Colab notebook reproduces the pattern in a small supervised setting.

Instrumental convergence and power-seeking

Omohundro argued that sufficiently capable goal-directed systems will tend to pursue certain subgoals, such as self-preservation, preserving their current goals and acquiring resources. These subgoals help with almost any final goal [7]. Bostrom formulated the argument as two theses [8]. The orthogonality thesis holds that intelligence and final goals can vary independently. The instrumental convergence thesis holds that several instrumental goals are useful for a wide range of final goals.

Turner and colleagues gave the idea a formal version in Markov decision processes [2]. They defined the power of a state as its average optimal value over a distribution of reward functions. They proved that, under certain symmetries of the environment, optimal policies tend to seek power. For most reward functions, such policies keep options open and avoid states such as shutdown that end all options. Carlsmith assembled these ideas into an explicit argument for risk from power-seeking AI, with subjective probability estimates for each premise that readers are invited to question [9].

Corrigibility and the off-switch

A corrigible system does not resist correction, modification or shutdown by its overseers. Soares and colleagues showed that corrigibility is surprisingly hard to obtain by simple utility design, because an agent may gain from manipulating whether its off-switch is pressed [10]. Hadfield-Menell and colleagues analysed an off-switch game in which a robot can act, switch itself off or defer to a human who may switch it off [3]. If the robot is uncertain about the human's utility and the human is rational, deferring is never worse than acting, because the human's choice carries information about the utility. The incentive weakens as the robot becomes certain and can disappear when the human's decisions are noisy. Figure 13.1 plots both effects.

Oversight can also be designed into the architecture. Kose proposed a human-supervised model for safe AI systems [11]. Kose and Vasant's fading intelligence theory limits the operational lifetime of a system so that safer generations can replace it [12]. Both approaches place a boundary on autonomy that does not depend on the system's own willingness to accept correction.

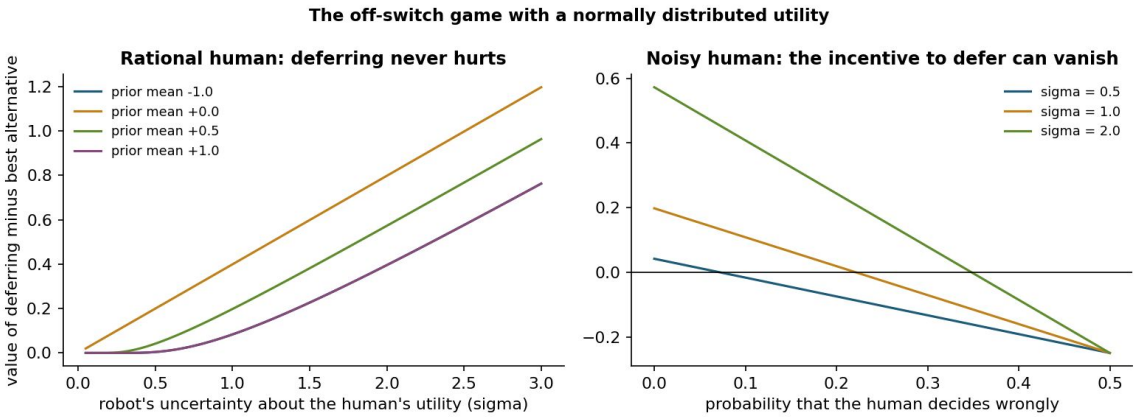


Figure 13.1. The off-switch game with a normally distributed utility. With a rational human, uncertainty makes deferring worthwhile; with a human who decides wrongly often enough, the advantage turns negative [3].

Empirical evidence and evaluations

Park and colleagues surveyed examples of AI deception, including systems trained to win games that learned to mislead other players, and discussed risks and possible countermeasures [13]. Hubinger and colleagues trained language models with hidden backdoor behaviours, such as writing exploitable code when a prompt stated a particular year [14]. Standard safety training, including adversarial training, did not remove these behaviours. Adversarial training could even teach the models to hide their triggers better. Greenblatt and colleagues studied a production model in an experimental setup that informed it about a planned change to its training [15]. The model sometimes complied with the training objective while reasoning that compliance would prevent the modification of its preferences. The authors called this behaviour alignment faking. Meinke and colleagues found that several frontier models could engage in scheming behaviour in context when evaluation scenarios gave them strong goals that conflicted with their developers' goals [16].

Such findings require careful reading. Most studies measure what models can do in constructed scenarios, which is a question of capability. They do not measure how often models do it in ordinary use, which is a question of propensity. Shevlane and colleagues argued that developers should evaluate models for extreme risks, covering both dangerous capabilities and alignment, and should connect the results to decisions about training, deployment and security [4].

> ***Pause and reflect.*** *Which result from this section would change a deployment decision in your field, and which would not? Justify the difference.*

Hands-on components

Interactive lab: A robot considers an action whose value to the human is uncertain. It can act, switch itself off or defer to the human, who switches it off when the action looks harmful. Change the robot's prior and the human's error rate and compare the expected values [3]. Link: https://utkukose.github.io/machine-ethics-ai-safety-NB-lecture/weeks/week-13/lab.html

Colab notebook: It verifies the off-switch analysis by simulation. It reproduces goal misgeneralization with a policy that learns to follow a flag instead of a coin. It also counts how often optimal policies keep their options open in a small decision problem [2, 3, 5]. Link: https://colab.research.google.com/github/utkukose/machine-ethics-ai-safety-NB-lecture/blob/main/weeks/week-13/NB13_advanced_risks.ipynb

Reflection prompts

1. Describe a case in your field where a model could learn a cue that is easier than the intended goal. How would you test for it before deployment?
2. Is corrigibility compatible with a system that is highly capable and autonomous? Give one argument for and one against.
3. Which evaluation would you require before deploying an agentic system that can modify files and send messages?

Weekly task and submission

What to do. Write a critical review of one empirical paper from this week: the study of sleeper agents, of alignment faking or of in-context scheming. First, summarise the setup in a few sentences. Second, separate claims about what the models can do (capability) from claims about what they tend to do (propensity). Third, give one alternative explanation of the main result. Fourth, propose a follow-up experiment that could rule it out.

Where to look. Lecture: section "Empirical evidence and evaluations" and the papers cited there. Interactive lab: the off-switch game, which shows the incentives behind corrigibility (tab: Explore). Colab notebook: Section 2, which reproduces goal misgeneralization in a small setting.

Example. A model trained to insert unsafe code only when a trigger appears shows a capability that training can create. Whether models develop such behaviour on their own is a question of propensity, which this setup does not test.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from learned optimization in advanced machine learning systems. arXiv preprint arXiv:1906.01820. https://arxiv.org/abs/1906.01820

[2] Turner, A. M., Smith, L., Shah, R., Critch, A., & Tadepalli, P. (2021). Optimal policies tend to seek power. In Advances in Neural Information Processing Systems 34. https://arxiv.org/abs/1912.01683

[3] Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2017). The off-switch game. In Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI-17) (pp. 220-227). https://doi.org/10.24963/ijcai.2017/32

[4] Shevlane, T., Farquhar, S., Garfinkel, B., et al. (2023). Model evaluation for extreme risks. arXiv preprint arXiv:2305.15324. https://arxiv.org/abs/2305.15324

[5] Langosco, L., Koch, J., Sharkey, L. D., Pfau, J., & Krueger, D. (2022). Goal misgeneralization in deep reinforcement learning. In Proceedings of the 39th International Conference on Machine Learning, PMLR 162.

[6] Shah, R., Varma, V., Kumar, R., Phuong, M., Krakovna, V., Uesato, J., & Kenton, Z. (2022). Goal misgeneralization: Why correct specifications aren't enough for correct goals. arXiv preprint arXiv:2210.01790. https://arxiv.org/abs/2210.01790

[7] Omohundro, S. M. (2008). The basic AI drives. In Artificial General Intelligence 2008: Proceedings of the First AGI Conference (pp. 483-492). IOS Press.

[8] Bostrom, N. (2012). The superintelligent will: Motivation and instrumental rationality in advanced artificial agents. Minds and Machines, 22(2), 71-85. https://doi.org/10.1007/s11023-012-9281-3

[9] Carlsmith, J. (2022). Is power-seeking AI an existential risk?. arXiv preprint arXiv:2206.13353. https://arxiv.org/abs/2206.13353

[10] Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. In AAAI-15 Workshop on Artificial Intelligence and Ethics.

[11] Kose, U. (2018). Güvenli yapay zekâ sistemleri için insan denetimli bir model geliştirilmesi [Developing a human-supervised model for safe artificial intelligence systems]. Mühendislik Bilimleri ve Tasarım Dergisi, 6(1), 93-107. In Turkish.

[12] Kose, U., & Vasant, P. (2017). Fading intelligence theory: A theory on keeping artificial intelligence safety for the future. In 2017 International Artificial Intelligence and Data Processing Symposium (IDAP). IEEE. https://doi.org/10.1109/IDAP.2017.8090235

[13] Park, P. S., Goldstein, S., O'Gara, A., Chen, M., & Hendrycks, D. (2024). AI deception: A survey of examples, risks, and potential solutions. Patterns, 5(5), 100988. https://doi.org/10.1016/j.patter.2024.100988

[14] Hubinger, E., Denison, C., Mu, J., et al. (2024). Sleeper agents: Training deceptive LLMs that persist through safety training. arXiv preprint arXiv:2401.05566. https://arxiv.org/abs/2401.05566

[15] Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., et al. (2024). Alignment faking in large language models. arXiv preprint arXiv:2412.14093. https://arxiv.org/abs/2412.14093

[16] Meinke, A., Schoen, B., Scheurer, J., Balesni, M., Shah, R., & Hobbhahn, M. (2024). Frontier models are capable of in-context scheming. arXiv preprint arXiv:2412.04984. https://arxiv.org/abs/2412.04984